



REMOTE PHOTOPLETHYSMOGRAPHY ESTIMATION

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ABSTRACT

Heart rate variability (HRV) is a vital indicator of health, stress, and emotional states, reflecting the balance between sympathetic and parasympathetic nervous system activity. Photoplethysmography (PPG) is a non-invasive optical technique to monitor cardiovascular function. Remote photoplethysmography (RPPG) has emerged as a remote heart rate monitoring tool without physical sensors, utilizing high-resolution camera recordings. Remote photoplethysmography (RPPG) estimates vital signs by measuring changes in the reflected light from the human skin. Compared to traditional PPG techniques, remote PPG enables contactless measurement at a reduced cost. The HRV has great significance in health assessment and emotional monitoring, emphasizing the role of RPPG in diverse applications such as health-care monitoring, stress management, and physical activity tracking.

Keywords: Heart rate variability (HRV), Remote photoplethysmography (rPPG), remote heart rate estimation

DEVICES FOR PPG



Figure 4: Various devices used for PPG [3].

REFERENCES

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STEPS FOR RPPG

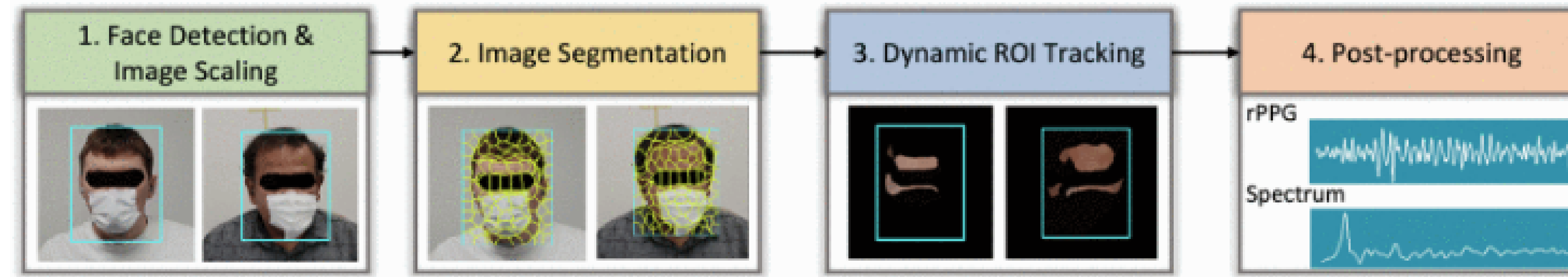


Figure 2: Basic Steps involved in RPPG [1].

MODEL

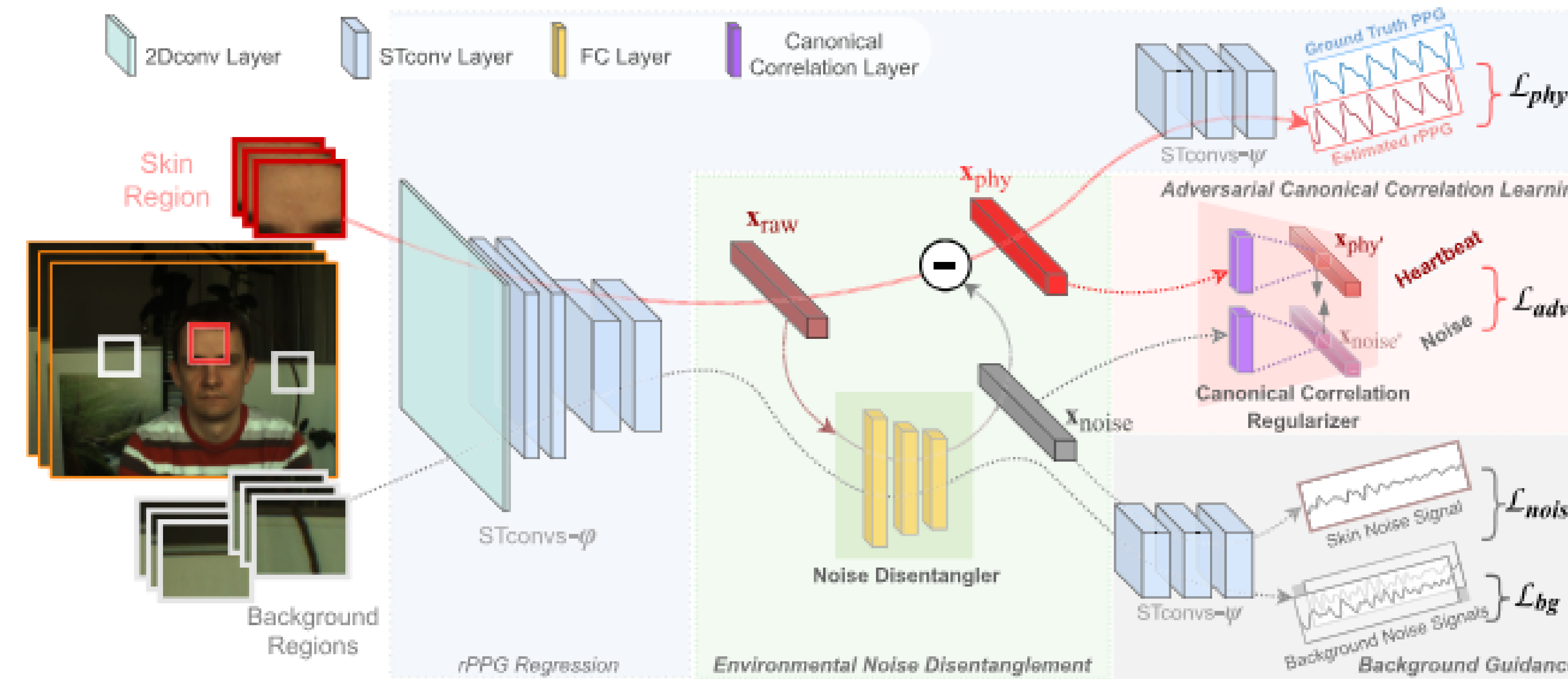


Figure 3: Block diagram of ND-DeeprPPG with noise disentanglement [2].

SUBJECTIVE EVALUATION

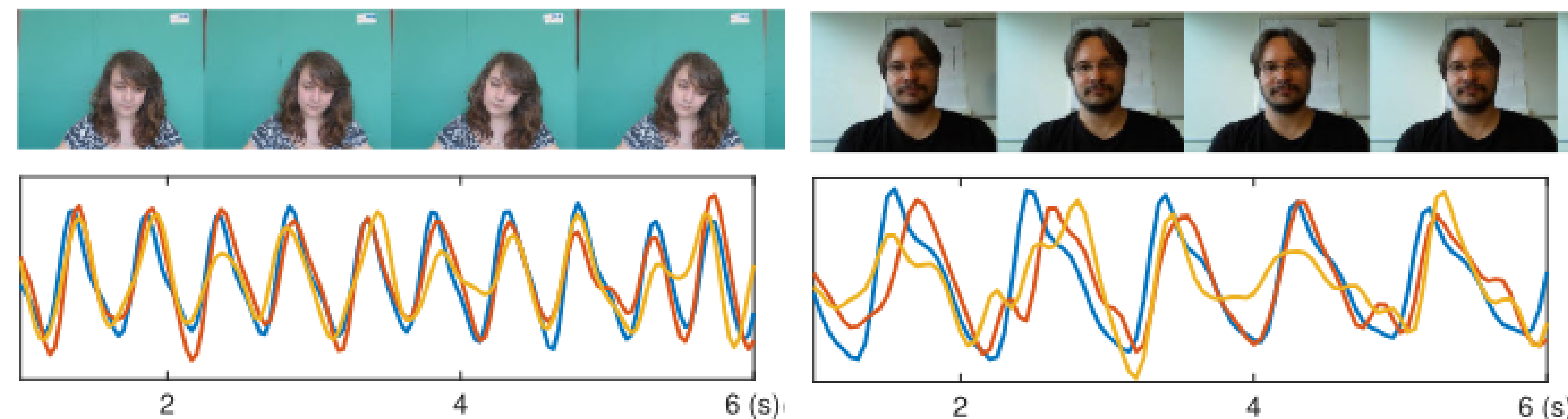


Figure 5: Qualitative comparisons of ND-DeeprPPG on COHFACE and UBFC.

OPTICAL HEART RATE SENSING

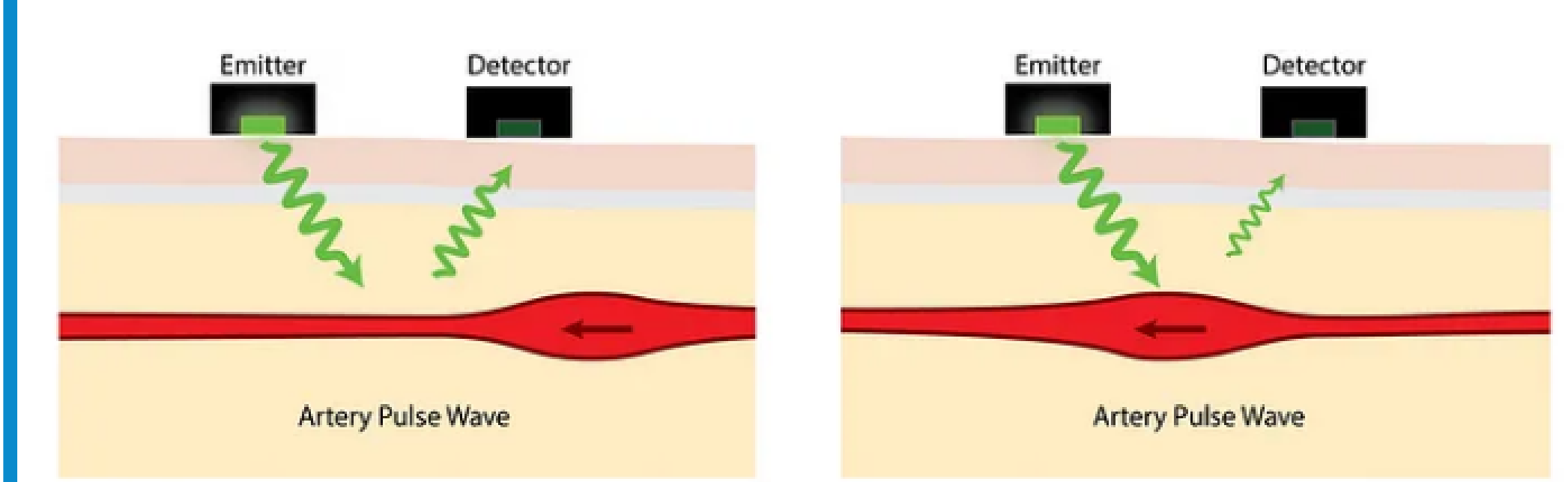


Figure 1: Optical heart rate sensing. Left: Low pressure, less absorption, high reflectivity. Right: High pressure, high absorption, low reflectivity.

OBJECTIVE EVALUATION

Table 1: Quantitive comparison of different methods for PURE and COHFACE dataset.

Methods	PURE		COHFACE	
	MAE ↓	RMSE ↓	MAE ↓	RMSE ↓
CHROM	2.07	2.50	8.44	13.74
HR-CNN	1.84	2.37	8.10	10.78
ETA-rPPGNet	0.34	0.77	4.07	6.65
CVD	1.29	2.01	14.5	16.5
Gideon2021	2.1	2.6	1.5	4.6
DeeprPPG	0.28	0.43	3.07	7.06
ND-DeeprPPG	0.18	0.41	0.64	1.89

APPLICATIONS

- Measuring multiple vital signs
- Affective computing
- Deepfake detection
- Face anti-spoofing

CONCLUSION

RPPG offers a promising avenue for comprehensive health monitoring and emotional assessment. The ability to remotely measure heart rate and assess autonomic nervous system activity without physical sensors provides convenience and accessibility in various domains such as healthcare, wellness, and human-computer interaction. Though it has many advantages over the traditional sensor-based approach, it has some cons, like motion artifacts and environmental factors like lighting conditions and temperature, which can affect the result of PPG.